Tutorial Week 4 – Search

Aim

- Review definitions.
- Experiment with BFS, DFS, A* algorithms by drawing search trees.
- Compare the performance of each of the Algorithms using the provided notebook.

1. **Define in your own words the following terms: state, state space, search tree, search node, goal, action, transition model, and branching factor.**

2. **Consider the Romania map studied in the lecture and reproduced below. The goal state is Bucharest. Calculate and show working for which path the A* algorithm will choose when:**
    a. **Starting from Oradea**
    b. **Starting from Sibiu**
    c. **Starting from Timisoara**
    d. **Starting from Arad, and currently in Sibiu**

3. **A*, BFS, DFS algorithm comparison.**
    a. Download Tut4search.ipynb
    b. Start jupyter notebook (or jupyter lab)
    c. Work through the notebook.

With reference to the Romania map below, work through Tut4search.ipynb.

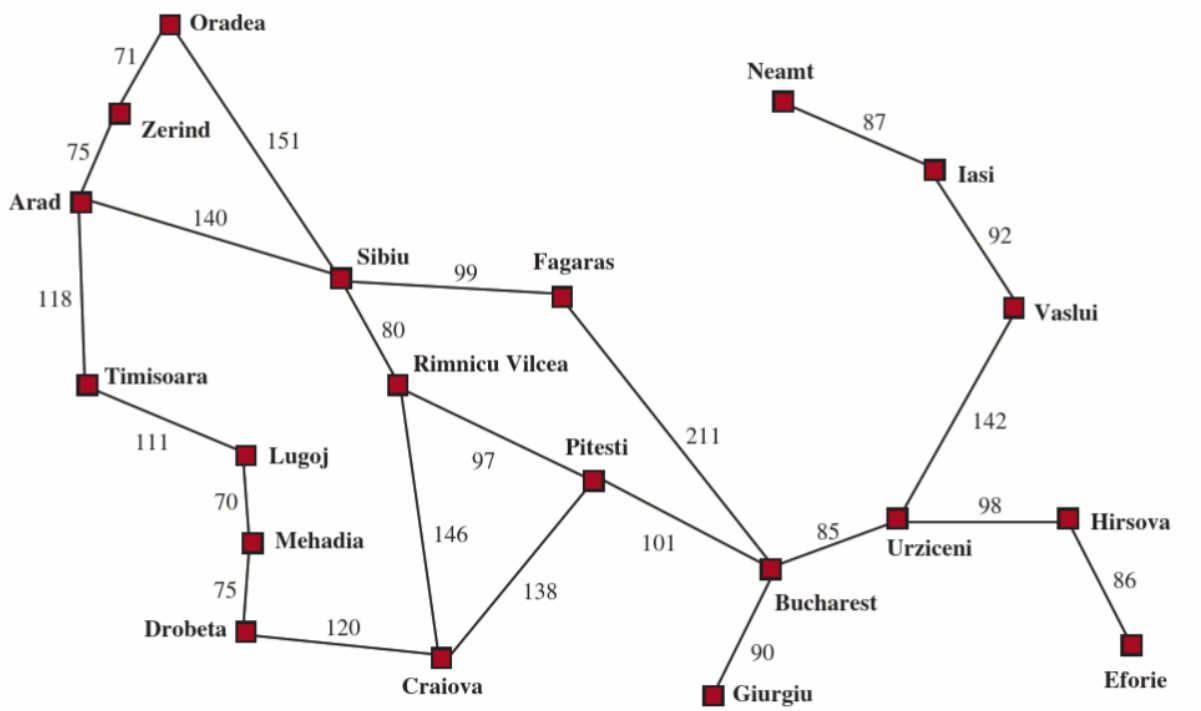


Figure 3.1 A simplified road map of part of Romania, with road distances in miles.

End